Never let browsers hold you back!

>> Modernizr helps you detect native support for HTML5 and CSS3 features in web browsers. We'll tell you why and how you could use it in your own projects! > by Ketan Singh

The advent of HTML5 and CSS3 and the ever-growing number of platforms trying to keep up with their standards has invoked unbridled enthusiasm among designers and developers alike. However, not many of them confidently use HTML5 and CSS3 at the production level, without any fallback options. This is primarily because of the slow transition to new and more supportive browsers by the users.

For designers, it has always been a pain to let go of those cool features available in newer web technologies, simply because some browsers don't support them as expected!

Nevertheless, if you do feel the urge to show your creativity with HTML5 and CSS3, while ensuring that the users with older browsers don't get left too far behind, Modernizr is here for you.

What is Modernizr?

Modernizr is a small JavaScript library that detects the availability of native support for individual CSS3 and HTML5 features in a web browser.

Although Modernizr sounds like it's out to 'modernize' every browser it runs on; that's not entirely true.

Its primary job is to detect support for individual features for you, which you as the developer have to make use of. However, it does have some 'modernizing powers' as far as Internet Explorer is concerned. Keep reading to find more on that.

What other options do I have?

Unfortunately, there is no such perfect tool or trick, which can solve all your cross-browser compatibility issues automatically. However, there are a number of hacks, tools and design decisions that can help you partially mitigate certain issues.

Most developers usually deploy the 'Browser sniffer' hack (user agent sniffing). They include a special JavaScript or PHP based snippet, called a 'sniffer' in a particular webpage, which is usually meant to detect the client's 'user agent' information.

The 'Browser sniffer' hack is very unreliable and ambiguous, because a browser's 'user agent' information is user-configurable.. Moreover, Modernizr was particularly built to replace 'browser sniffing' altogether, along with all the deception and ambiguity that comes with it.

A relatively saner decision would be to recognize a reasonable set of browsers that you are willing to support and then develop accordingly. This approach is widely used and you should consider using it if all the contents of your webpage are equally important and a strictly identical layout is expected across all the targeted browsers. However, it most probably eliminates any chance of using HTML5 and CSS3 features on the page.

Some developers opt for the graceful degradation design decision, which refers to the practice of building your webpage with the latest features and best browsers in mind. In addition, you add tools to detect the availability of support for those features. To prepare for 'no support,' you deploy alternative measures to scale down, but make sure a reasonable version is rendered.

Now that you have Modernizr in your arsenal, we'll tell you how it goes one-up!

What makes Modernizr so special?

Modernizr is a great tool when it comes to solving cross-browser compatibility issues, because of its power of feature detection. It is much more reliable and robust as compared to browser sniffing or any other hack. Browser sniffing means simply asking the browser a naïve "Who are you?".

Feature detection means telling the browser "Let me see what features you can support"! Now, you can't hide much, if your mom herself starts looking out for the candies in your pocket, can you? Modernizr expects you to understand that it is not always necessary to display things in an identical manner across every browser out there.

Modernizr is built upon the concept of progressive enhancement, which advocates rendering web pages progressively, based on characteristics of the client. It encourages you to build your projects in a bottom-up and layered way, in order to render individual 'feature-specific' layout and content on the fly. Modernizr expects you to understand that it is not always necessary to display things in an identical manner across every browser out there.

Developers use Modernizr mostly to accomplish two common motives. Some developers liberally use CSS3 in their projects, and use Modernizr to create 'hackish' CSS fallbacks to mimic the same effect in 'non-supportive' browsers.

Others use it to render reasonably distinct content, layout and behaviour, based on the support available for a particular set of individual features in the browser.

How do I install Modernizr in a webpage?

To start using Modernizr in your projects, you first need to 'build and download' your preferred version of Modernizr from http://